Mathematics Skills for Computing

2nd September 2024

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

SECTION A - Multiple Choice Question
Circle ONE (1) correct answer from A, B, C, or D for each question.
Each question is worth 1 mark.

Question 1

Simplify the following expression:
$p^8 \div p^5 \div p^{-2}$

A $\boldsymbol{p}$

B $\boldsymbol{p^5}$

C $\boldsymbol{p^{15}}$

D $\boldsymbol{p^{11}}$

1 mark

Question 2

Simplify the following expression:
$a^3 + 2a^6 + 4a^3 + a$

A $2a^6 + 5a^3 + a$

B $\boldsymbol{6a^{13}}$

C $\boldsymbol{6a^{12} + a}$

D $\boldsymbol{8a^9 + a^3 + a}$

1 mark

Question 3

Simplify the expression:
$(2t^4)^3$

A $8t^7$

B $\boldsymbol{2t^{12}}$

C $\boldsymbol{2t^7}$

D $\boldsymbol{8t^{12}}$

1 mark

© NCC Education Limited 2024

Marks

Question 4

Simplify the expression:

$\frac{16d^{15}}{8d^5}$

A $8d^3$

B $\mathbf{2d^3}$

C $\mathbf{8d^{10}}$

D $\mathbf{2d^{10}}$

1 mark

Question 5

Factorise $x^2 - x - 12$

A $(x + 3)(x - 4)$

B $\mathbf{(x + 3)(x + 4)}$

C $(x - 3)(x - 4)$

D $\mathbf{(x - 3)(x + 4)}$

1 mark

Question 6

Simplify $\frac{25x^4y}{5y}$

A $20x^4$

B $\mathbf{(5x^3 - y)(5x + y)}$

C $\mathbf{5x^4}$

D $\mathbf{5y^4}$

1 mark

Question 7

Simplify $\frac{23}{b^0}$

A $\frac{23}{b}$

B 22

C 23

D $\mathbf{23b}$

1 mark

© NCC Education Limited 2024

Marks

Question 8

Factorise completely $2\mathrm{v}^8 + 22\mathrm{v}^3\mathrm{w}^2$

A $2v(v^7 + 11v^2w^2)$

B $\boldsymbol{2(v^8 + 11v^8w^2)}$

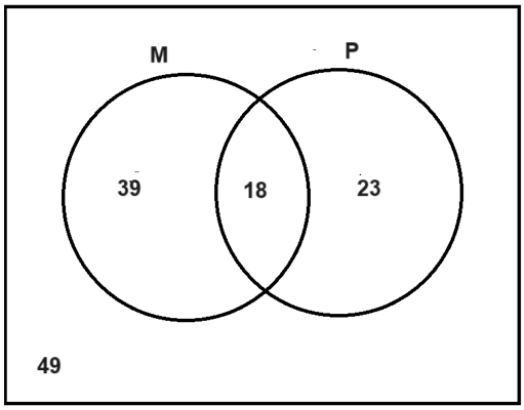
C $\boldsymbol{2(v^8 + 11v^3w^2)}$

D $\boldsymbol{2v^3(v^5 + 11w^2)}$

1 mark

Question 9

A group of students are surveyed about the subjects they are studying. The responses are displayed in the Venn diagram below where M represents Maths and P represents Physics.



Find the total number of students who were surveyed.

A 49

B 129

C 111

D 80

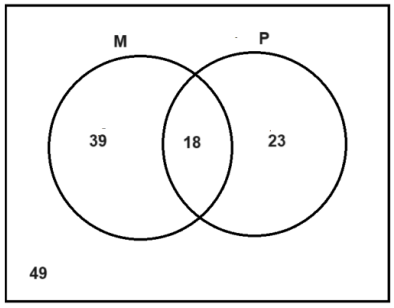
1 mark

© NCC Education Limited 2024

Marks

Question 10

Using the same Venn diagram as Question 9



Find $n\,(M \cup P)$

A 18

B 82

C 49

D 80

1 mark

Question 11

If C = {3, 7, 8, 9, 11, 14, 17, 28} find n(C)

A 28

B 8

C {3, 28}

D 10

1 mark

Question 12

An example of an empty set is:

A The set of prime numbers greater than ONE THOUSAND AND NINE (1009)

B The set of squares with THREE (3) sides

C The set of even numbers less than SEVEN (7)

D The set of numbers divisible by TEN (10)

1 mark

© NCC Education Limited 2024

Marks

Question 13

If D = {2, 5, 7, 8, 11} and E = {3, 5, 6, 8, 9, 12} find D ∩ E

A {5, 8}

B {2, 3, 5, 6, 7, 8, 9, 11, 12}

C {2, 3, 6, 7, 11, 12}

D {2, 3, 5, 5, 6, 7, 8, 8, 9, 11, 12}

1 mark

Question 14

Find the determinant of the matrix $\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$

A -6

B -10

C 5

D 2

1 mark

Question 15

Name the type of matrix shown here:

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 1 & 1 \end{pmatrix}$$

A Scalar matrix

B Identity matrix

C Triangular matrix

D Null matrix

1 mark

Question 16

What is the order of the following matrix?

$$\begin{pmatrix} 3 & 7 & 3 \\ 5 & 4 & 1 \end{pmatrix}$$

A 2 × 3

B 6

C 3 × 2

D 1

1 mark

© NCC Education Limited 2024

Marks

Question 17

Find the transpose of the following matrix

$$\begin{pmatrix} 4 & 6 \\ 2 & 9 \\ 5 & 1 \end{pmatrix}$$

A $\begin{pmatrix} 6 & 4 \\ 9 & 2 \\ 1 & 5 \end{pmatrix}$

B $\begin{pmatrix} \mathbf{4} & \mathbf{2} & \mathbf{5} \\ \mathbf{6} & \mathbf{9} & \mathbf{1} \end{pmatrix}$

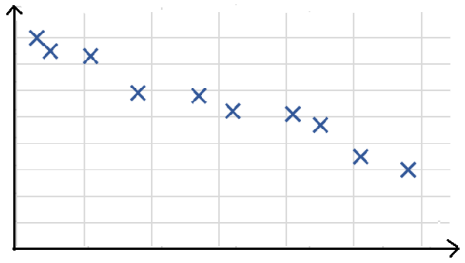
C $\begin{pmatrix} \mathbf{6} & \mathbf{9} & \mathbf{1} \\ \mathbf{4} & \mathbf{2} & \mathbf{5} \end{pmatrix}$

D $\begin{pmatrix} \mathbf{6} & \mathbf{4} \\ \mathbf{1} & \mathbf{5} \\ \mathbf{2} & \mathbf{9} \end{pmatrix}$

1 mark

Question 18

What type of correlation is show in the scatter graph below?



A No correlation

B Weak negative correlation

C Strong negative correlation

D Strong positive correlation

1 mark

Question 19

On which type of graph or chart is frequency density plotted?

A Pie chart

B Bar chart

C Histogram

D Boxplot

1 mark

© NCC Education Limited 2024

Marks

Question 20

200 people were asked to state their preferred brand of toothpaste. The results are below.

	Whiter Smile	Minty Fresh	Minton	Bright Smile
Frequency	92	12	30	66

A pie chart of the data is drawn to illustrate the results. Calculate the angle of the sector for Minton toothpaste.

A 30°

B 15°

C 54°

D 0.15°

1 mark

Question 21

A data set is given by M = {15, 13, 11, 14, 13, 14, 10, 12, 15, 14}
Calculate the mean of the data set.

A 10

B 13.1

C 5

D 13.5

1 mark

Question 22

A data set is given by M = {15, 13, 11, 14, 13, 14, 10, 12, 15, 14}
Calculate the mode of the data set.

A 15

B 13.1

C 14

D 5

1 mark

© NCC Education Limited 2024

Marks

Question 23

A data set is given by M = {15, 13, 11, 14, 13, 14, 10, 12, 15, 14}
Calculate the median of the data set.

A 14

B 13.1

C 13.5

D 5

1 mark

Question 24

The standard deviation of a set of data is given as 16. What is the variance?

A 256

B 0.16

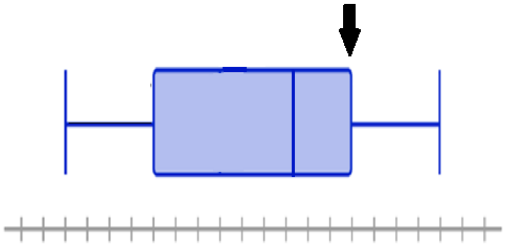
C 4

D 32

1 mark

Question 25

In the boxplot below, what is the name of the statistical measure which the arrow is pointing to?



A Interquartile range

B Median

C Upper quartile

D Lower quartile

1 mark

© NCC Education Limited 2024

Marks

Question 26

In an experiment, a five-sided spinner, numbered 1 to 5 is spun 120 times. The results are recorded in the table below.

Score	1	2	3	4	5
Frequency	15	34	14	18	39

What is the relative frequency of the spinner **not** landing on a 2? Express your answer as a fraction in its simplest form:

A $\frac{120}{86}$

B $\mathbf{\frac{34}{120}}$

C $\mathbf{\frac{17}{60}}$

D $\mathbf{\frac{43}{60}}$

1 mark

Question 27

Jo wears a suit, shirt and tie for work. He has THREE (3) ties, FIVE (5) suits and TEN (10) shirts. How many different work outfits can be selected from these items?

A 18

B 10

C 125

D 150

1 mark

Question 28

The probability that Jo takes the train to work is 0.7. What is the probability that Jo does **not** take the train to work?

A 0.3

B 0.7

C 0.49

D 1.4

1 mark

© NCC Education Limited 2024

Marks

Question 29

The probability that the 7:20 am train will be delayed is 0.1
What is the probability that the 7.20am train will be delayed on Monday morning **and** on Tuesday morning?

A 0.1

B 0.9

C 0.2

D 0.01

1 mark

Question 30

A fair, six-sided dice is rolled. What is the theoretical probability of the dice landing on a FIVE (5)?

A $\frac{1}{5}$

B $\frac{\mathbf{5}}{\mathbf{6}}$

C $\frac{\mathbf{1}}{\mathbf{6}}$

D $\frac{\mathbf{1}}{\mathbf{2}}$

1 mark

© NCC Education Limited 2024

Marks

SECTION B
ANSWER ALL QUESTIONS

You must show your workings.
Marks are awarded for these.

© NCC Education Limited 2024

Marks

Section B

Question 31

a) Solve the following quadratic equation by using the Quadratic Formula: **2**

$4x^2 - 6x + 1 = 0$

Show your working. Leave your answer in surd form.

b) Solve the simultaneous equation $2x + 3y = 17$ and $8x + y = 13$ algebraically and find the values of x and y. Show your working. **3**

c) Transpose the following formula to make b the subject. Show your working. **2**

$$a = \frac{3 - b}{8c + 2}$$

d) Simone goes to a café and buys TWO (2) cups of tea and THREE (3) hot chocolates. A hot chocolate costs 1.5 times the cost of a cup of tea. She receives £9.85 change from £30. Calculate the cost of a cup of tea. **3**

Total 10 Marks

Question 32

a) Show that $y = -2x + 3$ and $4y + 8x = 2$ are parallel to each other. **2**

b) Copy and complete the table of values for the curve $y = 3x^2 + 6x - 7$ for $-4 \leq x \leq 2$ **3**

x	-4	-3	-2	-1	0	1	2
$y = 3x^2 + 6x - 7$							

c) Use the values from (b) to plot a graph of the curve $y = 3x^2 + 6x - 7$ You should use the graph paper provided. **4**

d) Use the graph constructed in part (c) to solve the equation $3x^2 + 6x - 7 = 0$ **1**

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 33

a) If *A* is the set {5, 9, 11} and *B* is the set {1, 5, 9, 11, 18, 19}

i) State whether $A \subset B$. Explain your answer. **2**

ii) Find $B - A$. **1**

iii) Find $n(B - A)$. **1**

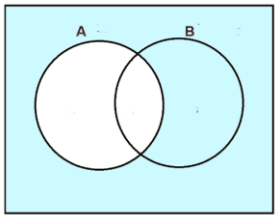
b) THIRTY-EIGHT (38) students were asked about which sports they like. **4**

- FOURTEEN (14) like football
- TWENTY-ONE (21) like cricket
- SIX (6) like both football and cricket.

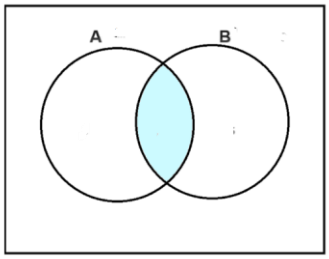
Draw a Venn diagram to represent this data.

c) State the set notation for the shaded areas of each of the following Venn diagrams.

i) **1**



ii) **1**



Total 10 Marks

© NCC Education Limited 2024

Marks

Question 34

a) TWO (2) matrices are given as
$\boldsymbol{D} = \begin{pmatrix} 3 & -2 \\ 7 & 4 \end{pmatrix}$ and $\boldsymbol{E} = \begin{pmatrix} -4 & 3 \\ 2 & 1 \end{pmatrix}$

i) Calculate $\boldsymbol{D} + \boldsymbol{E}$ **1**

ii) Calculate $\boldsymbol{D} - \boldsymbol{E}$ **1**

iii) Calculate $-\mathbf{2}\boldsymbol{D}$ **1**

b) TWO (2) matrices are given as **1**
$\boldsymbol{J} = \begin{pmatrix} 2 & 3 \\ -4 & 0 \end{pmatrix}$ and $\boldsymbol{K} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$
Calculate $\boldsymbol{JK}$

c) Given that **3**

$$\boldsymbol{P} = \begin{pmatrix} 1 & 2 \\ -3 & 0 \end{pmatrix}$$

show that P has an inverse and find $\boldsymbol{P^{-1}}$

d) TWO (2) matrices are given as **3**
$\boldsymbol{V} = \begin{pmatrix} 4 & 2 \\ -1 & 5 \end{pmatrix}$ and $\boldsymbol{W} = \begin{pmatrix} 1 & 6 \\ 2 & 1 \end{pmatrix}$

Find the product $\boldsymbol{VW}$ and then apply it to the coordinate vector $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 35

The distance (in miles) that a group of students travel to college is recorded in the table below.

Distance, d (miles)	Frequency
$0 \leq d < 1.0$	3
$1.0 \leq d < 2.0$	7
$2.0 \leq d < 3.0$	15
$3.0 \leq d < 4.0$	18
$4.0 \leq d < 5.0$	13
$5.0 \leq d < 6.0$	3
$6.0 \leq d < 7.0$	1

a) State whether the data is continuous or discrete. **1**

b) Calculate the cumulative frequency of the data. **1**

c) Present the data as a cumulative frequency curve (Ogive). Use the graph paper provided. **5**

d) Use your cumulative frequency graph from part (c) to estimate the value of the upper quartile. You must show how you have found the upper quartile on your graph **2**

e) The lower quartile is estimated to be 2.4 miles. Calculate an estimate of the interquartile range. **1**

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 36

a) 260 children visit an aquarium. The ages (years) of the children are shown in the table below.

Age, a (years)	Frequency
$0 \leq a < 2$	18
$2 \leq a < 4$	27
$4 \leq a < 6$	28
$6 \leq a < 8$	46
$8 \leq a < 10$	53
$10 \leq a < 12$	39
$12 \leq a < 14$	31
$14 \leq a < 16$	18

i) Calculate an estimate of the mean of the data. State the level of accuracy to which you have rounded your answer. **4**

ii) State within which class interval the modal value will lie and then calculate an estimate of the value using the formula for the mode. State the level of accuracy to which you have rounded your answer. **3**

iii) State within which class interval the median value will lie and then calculate an estimate of the value using the formula for the median. State the level of accuracy to which you have rounded your answer. **3**

Total 10 Marks

© NCC Education Limited 2024

Marks

Question 37

The probability of it raining on Monday is $\frac{2}{3}$

The probability of it raining on Tuesday is $\frac{4}{5}$

Assume that the two outcomes are independent.

a) Draw a *probability tree diagram* to show this information and all the possible outcomes. You should state all possible outcomes **and** calculate the probability of each outcome. **8**

b) Calculate the probability that it will rain on **one** day **only.** **2**

Total 10 Marks

End of Paper

© NCC Education Limited 2024

Foundation Maths Formula Sheet.

Quadratic equations

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Mean

Ungrouped data:

$$\bar{x} = \frac{\sum x}{n}$$

Ungrouped frequency table:

$$\bar{x} = \frac{\sum fx}{n} = \frac{\sum fx}{\sum f}$$

Grouped frequency table:

$$\bar{x} = \frac{\sum fm}{n} = \frac{\sum fm}{\sum f}$$

Mode

Grouped data

Estimated mode = $L + \frac{(F_m - F_b) \times c}{(2F_m - F_b) - F_n}$

Median

Grouped data

Estimated median = $L_{med} + W_{med} \times \frac{\frac{n}{2} - C_{med}}{F_{med}}$

Quartiles

Ungrouped data:

$$LQ = \frac{1}{4}(n+1)$$

$$UQ = \frac{3}{4}(n+1)$$

Grouped data

Quartile = $L + \frac{(Q-P) \times I}{F}$

Mean deviation

Ungrouped data

© NCC Education Limited 2024

Mean deviation $= \frac{\sum|x-\bar{x}|}{n}$

Grouped data
Mean deviation $= \frac{\sum f|x-\bar{x}|}{\sum f}$

Variance

Ungrouped data:
$$s^2 = \frac{\sum(x - \bar{x})^2}{n} = \frac{\sum x^2}{n} - \bar{x}^2$$

Grouped data:
$$s^2 = \frac{\sum f(x - \bar{x})^2}{\Sigma f} = \frac{\sum f x^2}{\Sigma f} - \left[\frac{\sum f x}{\Sigma f}\right]^2$$

Standard deviation

Ungrouped data:
$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Grouped data:
$$s = \sqrt{\frac{\sum f x^2}{\Sigma f} - \left[\frac{\sum f x}{\Sigma f}\right]^2}$$

Coefficient of Variation
$$CV = \frac{s}{\bar{x}}$$

Determinant of Matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

Inverse of Matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

© NCC Education Limited 2024